Shahrzad Dehghani

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Portfolio LinkedIn ORCiD GitHub ResearchGate

Profile

Researcher with experiences in simulations and data-driven analysis of practical problems using Machine learning approaches in academia and industry. Experiences in running and analyzing large-scale computational workloads on high-performance computing (HPC) systems, including the Jülich Supercomputing Centre (JSC) and Pleiades at the Bergische Universität Wuppertal. Interested in solving business-oriented problems using data.

Skills

Languages	English (C1), German (B1-B2), Persian (C2).
Programming languages	Python, MATLAB, SQL, C++
Machine Learning	Scikit-learn, TensorFlow, Keras, PyTorch
Software	LaTeX, Excel, Power BI, Tableau, OriginLab, XPSpeak41, SpecsLab Prodigy
High-Performance Computing (HPC)	Distributed CPU/GPU computing, Job scheduling (SLURM), Large-scale numerical simulations, data-intensive ML workloads
Version Control	Git, GitHub
Web Dev	HTML, CSS

Employment History

08.2023 – Present **Researcher in Optoelectrical Engineering at the Bergische Universität Wuppertal, Germany**

- Applied simulations and machine learning techniques to develop inverse design models for hybrid waveguide gratings using reflection spectra. Here are project explanation and paper.
- Research project conducted for Louisenthal Company on AI/data-driven optics research (Confidential project).
- Developed a GUI-based application for evaluation of university applicants. The App is currently used by the Bergische Universität Wuppertal.

11.2022 – 07.2023 **Internship at the Universität zu Köln, Germany**

- Thin film evaporation & fabrication by solution processing (Spin coating).
- UHV systems: for deposition and surface-sensitive spectroscopy (XPS, UPS, IPES, REELS).

9.2021 – 08.2022 **Research Assistant - Master Thesis - at the Forschungszentrum Jülich, Germany**

- Analyzed large-scale N-body numerical simulations of star clusters using NBODY6++ to study the dynamical evolution of star clusters. Investigated environmental effects on protoplanetary disc sizes. Considered different cluster scenarios, including varying star formation efficiency and gas removal duration, across clusters with different masses.

7.2020 – 10.2020 **Summer Internship at the Universität Bonn, Germany**

- Analyzed molecular line emission and hyperspectral image data using the SPy module python to study the physical, kinematic, and chemical properties of star-forming molecular clouds.

02.2018 – 12.2019 **Research Assistant at the Institute for Research in Fundamental Sciences (IPM) in Tehran, Iran**

- Processed photometric on observational data to compile a stellar catalog and estimate the star formation history, stellar mass, and distance of the SagDIG galaxy (INT survey)

Education

04.2020 – 03.2023 **M.Sc. in Physics, Universität zu Köln**.
Thesis title: *Role of gas expulsion time on cluster expansion. [Jülich Supercomputing Center (JSC)]*

10.2014 – 07.2016 **M.Sc. in Theoretical Physics, Alzahra University, Tehran, Iran**

10.2009 – 08.2014 **B.Sc. in Physics, Shahid Beheshti University, Tehran, Iran**
Specialization : Atomic and Molecular Physics

Independent Projects

03.2026 – Present **Mining insights from Amazon Jewelry Reviews**
From business analytics to LLM-driven insight generation - Data Analysis - Data Science
Goal: Analyze Amazon jewelry reviews to uncover customer satisfaction drivers, brand performance, and product issues using both data analytics and NLP/LLM techniques.

- **Part 1: Data Analysis (Business insights): brand benchmarking, price-segment analysis, and review-text mining.**
- Conducted exploratory data analysis (EDA) and NLP-based review text analysis on a large-scale Amazon jewelry dataset (6.8M reviews, 800k products, 23k brands) to uncover insights on product trends, brand performance, pricing, ratings, and customer sentiment.
- Applied regression analysis to quantify the impact of different complaint types (e.g. durability, appearance, etc.) on product ratings and identify key drivers of customer dissatisfaction across price segments.
- **Part 2: Data Science (NLP & Transformer-based modeling)**
- Developed a transformer-based sentiment classification using transformer models (DistilBERT, RoBERTa) on review data.
- In progress: Topic modeling, LLM-based summarization, aspect-based sentiment analysis (e.g. durability, design, size), early product success prediction, and detection of potentially fake reviews.
- The full analysis, codes, results, visualizations, and thorough explanations are available in my Github or my Portfolio.

Research Publications

1. S. Dehghani, C. Knoth, et al., "Data-driven inverse design of hybrid waveguide gratings using reflection spectra via tandem networks and conditional vaes," *MDPI - Optics*, Nov. 2025. DOI: 10.3390/opt6040061.
2. T. Parto, S. Dehghani, et al., "The isaac newton telescope monitoring survey of local group dwarf galaxies. v. the star formation history of sagittarius dwarf irregular galaxy derived from long-period variable stars," *The Astrophysical Journal*, Jan. 2023. DOI: 10.3847/1538-4357/aca471.
3. S. Pfalzner, S. Dehghani, et al., "Most planets might have more than 5 myr of time to form," *The Astrophysical Journal Letters*, Oct. 2022. DOI: 10.3847/2041-8213/ac9839.
4. T. Parto, S. Dehghani, et al., "Int monitoring survey of local group dwarf galaxies: Star formation history and chemical enrichment," *Communications of the Byurakan Astrophysical Observatory*, 2020, ISSN: 2579-2776. DOI: 10.52526/25792776-2020.67.2-232.

Volunteering

06.2017 - 08.2017 **Build Your Future** - Held Soft Skills Workshops (AIESEC in Baku, Azerbaijan)